

Degree-57 Moore graph: exact reductions, a global group obstruction, and the remaining gap

1. Result status

The target is a simple graph G with

$$\begin{aligned} (v,k,\lambda,\mu) &= (3250,57,0,1), \\ A^2 &= 57A - 56I + J. \end{aligned}$$

No complete construction or nonexistence certificate is supplied here. The strongest verified results are:

1. an exact equivalence with a $(57,56,1)$ distance-regular antipodal cover of K_{57} ;
2. an exact permutation-matching formulation;
3. a proof that **every** semiregular group-of-derangements construction is impossible, including all cyclic, abelian, and nonabelian groups of order 56;
4. a more general representation-theoretic restriction: every nontrivial irreducible constituent of the generated 56-point permutation action must have degree divisible by 5;
5. a compressed 3136-by-3136 zero-one matrix formulation with a forced spectrum and four exact matrix identities.

The unresolved case is a genuinely nonregular antipodal cover: a compatible family of 56-point matching permutations not contained in any semiregular group.

2. Full graph spectrum

On the all-one vector, A has eigenvalue 57. On its orthogonal complement, $J=0$, so

$$A^2 + A - 56I = 0.$$

The remaining eigenvalues are therefore 7 and -8. If their multiplicities are f, g , then

$$\begin{aligned} f+g &= 3249, \\ 57 + 7f - 8g &= \text{tr}(A)=0. \end{aligned}$$

Hence

$$\text{Spec}(A) = \{57^1, 7^{1729}, (-8)^{1520}\}.$$

This is feasible and gives no contradiction by itself.

3. Rooted decomposition

Fix a vertex u . Its neighborhood consists of 57 pairwise nonadjacent branch vertices $b_0, \dots, b_{56}$. Every remaining vertex is adjacent to exactly one branch vertex, because triangles and 4-cycles are forbidden. Thus the 3192 distance-two vertices split into fibres

$$X_i = N(b_i) \setminus \{u\}, \quad |X_i|=56.$$

Every vertex in X_i has exactly one neighbor in every X_j , $j \neq i$. Therefore each pair of distinct fibres is joined by a perfect matching.

Order the vertices as $(u, b_0, \dots, b_{56}, X_0, \dots, X_{56})$. Let R be the 57-by-3192 fibre incidence matrix and C the adjacency matrix induced on the distance-two vertices. Then

$$A = \begin{bmatrix} 0 & 1^T & 0 \\ 1 & 0 & R \\ 0 & R^T & C \end{bmatrix},$$

$$\begin{aligned} R R^T &= 56 I_{57}, \\ R^T R &= P := I_{57} \text{ tensor } J_{56}. \end{aligned}$$

Comparing blocks in $A^2 = 56I - A + J$ gives

$$\begin{aligned} C R^T &= J - R^T, \\ R C &= J - R, \\ C^2 &= 56 I - C + J - P. \end{aligned} \tag{1}$$

Conversely, a zero-one symmetric matrix C with zero diagonal, zero diagonal fibre blocks, permutation off-diagonal blocks, and equation (1) reconstructs the full Moore graph by adding u and the 57 branch vertices. This is an exact equivalence, not merely a necessary local condition.

3.1 Distance-regular antipodal cover

The graph with adjacency matrix C has degree 56. Two vertices in the same fibre are at distance 3; fibres are the antipodal classes. Its intersection array is $\{56, 55, 1; 1, 1, 56\}$.

Thus the degree-57 Moore problem is equivalent to the existence of a $(57, 56, 1)$ distance-regular antipodal cover of K_{57} (DRACKN).

3.2 Spectrum of the rooted cover

The fibre-constant subspace has quotient matrix $J_{57} - I_{57}$, hence eigenvalues 56 once and -1 with multiplicity 56. On $\ker R$, both J and P vanish, so (1) becomes

$$C^2 + C - 56I = 0.$$

The roots are 7 and -8. Trace zero gives

$$\text{Spec}(C) = \{56^1, 7^{1672}, (-1)^{56}, (-8)^{1463}\}.$$

4. Exact matching-permutation system

Label every fibre by the same 56-element set. Let P_{ij} be the permutation matrix of the matching from X_i to X_j ; then

$$\begin{aligned} P_{ii} &= I, \\ P_{ji} &= P_{ij}^T. \end{aligned}$$

Relabel each fibre independently so that, for one distinguished fibre X_0 ,

$$P_{0i} = P_{i0} = I_{56} \quad \text{for } i \neq 0. \quad (2)$$

Equation (1) becomes a finite collection of exact permutation identities. In particular, for every $i \neq 0$,

$$I + \sum_{j \neq 0, i} P_{ji} = J_{56}. \quad (3)$$

Hence the 56 permutations consisting of the identity and the 55 matchings in any normalized column form a sharply transitive set: for every ordered pair of labels (x, y) , exactly one of those permutations maps x to y .

For distinct i, j , the off-diagonal block equation is

$$\sum_{h \neq i, j} P_{ih} P_{hj} = J_{56} - P_{ij}. \quad (4)$$

Equations (2)-(4), together with transpose symmetry and permutation constraints, are an exact finite formulation. A solution reconstructs the graph; an exhaustive proof of unsatisfiability resolves nonexistence.

5. The all-group obstruction

Theorem

The matching system cannot be contained in a semiregular permutation group on the 56 labels. Consequently no regular group cover, cyclic or nonabelian, can yield the degree-57 Moore graph.

Proof

Assume all matching permutations belong to a semiregular group $G \leq S_{56}$. Every orbit of a semiregular action has size $|G|$, so $|G|$ divides 56. For a fixed normalized column, equation (3) contains 55 pairwise distinct nonidentity permutations: if two were equal, two vertices would acquire the same two common neighbors, producing a 4-cycle. Thus $|G| \geq 56$, whence

$$|G| = 56,$$

and the action is regular.

Every group of order 56 has a nontrivial one-dimensional complex character. Let n_7 be the number of Sylow 7-subgroups. Then n_7 is 1 or 8.

- If $n_7=1$, quotienting by the normal Sylow 7-subgroup gives a group of order 8, which has a quotient of order 2.
- If $n_7=8$, the eight Sylow 7-subgroups contribute $8 \cdot (7-1)=48$ distinct non-identity elements. The seven remaining nonidentity elements must constitute the unique Sylow 2-subgroup; it is normal, and the quotient has order 7.

In either case there is a nontrivial homomorphism to a cyclic group and hence a nontrivial linear character χ .

Because the action is regular, every 56-by-56 matching block is a regular permutation matrix $P(g_{ij})$. Apply the character χ to every group-algebra block in (1). The matrices J_{56} map to

$$\sum_{g \in G} \chi(g) = 0,$$

so both J and P disappear. We obtain a 57-by-57 Hermitian matrix S_χ satisfying

$$\begin{aligned} S_\chi^2 + S_\chi &= 56 I_{57}, \\ \text{tr}(S_\chi) &= 0. \end{aligned} \tag{5}$$

Its eigenvalues lie in $\{7, -8\}$. If their multiplicities are a, b , then

$$\begin{aligned} a+b &= 57, \\ 7a-8b &= 0. \end{aligned}$$

Thus $a=152/5$, impossible. This contradiction proves the theorem.

Independent corroboration

A regular deck group of order 56 contains an involution by Cauchy's theorem, and the deck action extends to an automorphism of the reconstructed Moore graph fixing the root and branch vertices. A 2026 preprint proves that a degree-57 Moore graph has no involutory automorphism. That gives a shorter, literature-dependent exclusion. Equation (5) is independent of that preprint and is the preferred certificate.

6. A universal representation constraint beyond regular covers

Let $\Gamma \leq S_{56}$ be the group generated by all matching permutations of an arbitrary solution. Connectivity makes the action transitive. Decompose its natural permutation module as

$$C^{56} = 1 \text{ direct_sum } U.$$

For every nontrivial irreducible constituent ρ of U , of degree d , apply ρ blockwise to (1). On a single copy of ρ this yields a $57d$ -dimensional Hermitian matrix with eigenvalues 7 and -8 and trace zero. Therefore their multiplicities are

$$\begin{aligned} m_7 &= 152d/5, \\ m_{-8} &= 133d/5. \end{aligned}$$

They must be integers, so

$$5 \text{ divides } d. \tag{6}$$

Thus **every nontrivial irreducible constituent appearing in the 56-point permutation module has dimension divisible by 5.**

This is a genuine restriction, but not a contradiction: for example, the standard constituent of the natural action of S_{56} has dimension 55. Therefore (6) does not eliminate the general nonregular case.

7. A compressed 3136-by-3136 formulation

Set $N=I+C$, normalize as in (2), and delete the rows and columns belonging to X_0 . Call the remaining matrix B . It is a symmetric zero-one matrix of order $56^2=3136$, partitioned into 56-by-56 blocks. Every block, including the diagonal blocks, is a permutation matrix.

Define

$$\begin{aligned} P &= I_{56} \text{ tensor } J_{56}, \\ Q &= J_{56} \text{ tensor } I_{56}, \\ J &= J_{3136}. \end{aligned}$$

Direct block elimination from (1) gives four exact identities:

$$B^2 - B = 56 I + J - P - Q, \tag{7}$$

$$BP = PB = J, \tag{8}$$

$$BQ = QB = J. \tag{9}$$

These identities encode both sets of parallel-class sums and are useful for exact SAT, SMT, or integer-programming formulations.

7.1 Forced spectrum of B

Decompose the space according to the two commuting rank-56 matrices P, Q .

- On the all-one vector, B has eigenvalue 56.
- On $(\text{im } P \cap \ker Q) \oplus (\text{im } Q \cap \ker P)$, dimension 110, equations (8)-(9) force eigenvalue 0.
- On $\ker P \cap \ker Q$, dimension $(56-1)^2=3025$, equation (7) reduces to $B^2-B=56I$, so the eigenvalues are 8 and -7.

Since $\text{tr}(B)=3136$, the remaining multiplicities are 1617 and 1408. Therefore

$\text{Spec}(\mathbf{B}) = \{56^1, 8^{1617}, 0^{110}, (-7)^{1408}\}.$

8. What was tested and what failed to close

The scripts in this package exactly test the full matrix equation, rooted equation, matching blocks, and compressed identities on the known Moore graphs of degrees 2, 3, and 7. This guards against a reduction that accidentally excludes Petersen or Hoffman-Singleton behavior.

The following routes did not produce a complete contradiction:

- basic spectrum and multiplicity feasibility;
- the ordinary Krein nonnegativity conditions for the antipodal cover;
- parity/sign arguments for arbitrary matching permutations;
- replacing the matching family by a group without first proving it is a group;
- the constituent-divisibility condition (6), because large transitive groups can satisfy it;
- local sharply-transitive/Latin-square constraints without a global classification theorem.

No claim is made that these routes are exhausted in all possible refinements.

9. Exact remaining proof obligation

A complete resolution now requires one of the following:

Existence

Construct 57 fibres of size 56 and permutation matrices P_{ij} satisfying (2)-(4), then verify all equations exactly and reconstruct the 3250-vertex graph.

Nonexistence

Prove that no **nonregular** family of matchings can satisfy (2)-(4). Equivalently, prove that no symmetric zero-one block matrix $\mathbf{B}$ with permutation blocks satisfies (7)-(9).

The group theorem closes an entire natural construction family, but the non-regular family is not a cosmetic leftover: it includes known Moore behavior at degree 7, so discarding it would be logically invalid.

10. Suggested next exact computation

The most direct proof-producing search target is $\mathbf{B}$, not the full 3250-by-3250 adjacency matrix:

1. treat the 56 diagonal blocks as fixed identities;
2. represent each off-diagonal block by a permutation variable;

3. impose symmetry and equations (7)-(9) incrementally;
4. quotient only by the rigorously available action `S_56 wr S_56` from relabeling fibres and labels, with canonical augmentation;
5. emit DRAT/LRAT or another independently checkable unsatisfiability certificate;
6. validate the encoder on the degree-3 and degree-7 instances before the degree-57 run.

This remains a very large exact search. The reduction supplied here lowers the order from 3250 to 3136 and, more importantly, replaces generic graph variables by structured permutation blocks and exact quadratic identities.

11. Literature checked

The external literature reviewed for status and comparison included:

- Faber and Keegan, *The Existence of a Moore Graph of Degree 57 Is Still Open* (2022), which identifies a gap in a claimed nonexistence proof.
- Smith and Montemanni, *The Moore Graph of Diameter 2 and Degree 57 via Cyclic Derangements* (2026), which excludes a cyclic derangement construction but leaves noncyclic and nongroup cases.
- Ishida, *No involutions in the missing Moore graph* (2026 preprint), which proves an automorphism restriction.
- Standard literature on distance-regular antipodal covers of complete graphs.

A targeted search did not locate the exact all-groups character argument in Section 5. This is not a definitive novelty claim; it records only the scope of the search performed.